



CCDS Final Year Project

Full Stack Development and Deployment of a
Web-Based Speech Recognition System

Project Plan

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Contents

1	Project Overview	3
2	Project Objectives	3
3	Features and Tasks	3
4	Project Schedule	5
5	Gantt Chart	6

Project Overview

MAGOR is a web-based audio and video transcription application that supports the upload and processing of recordings, utilizing advanced speech-to-text models for automatic speech recognition. MAGOR provides an array of features, including transcript generation and editing, captioning of audio and image content, speaker detection, and tagging of recordings.

Although MAGOR currently provides a functional transcription system, several components remain incomplete or unstable. This project intends to improve the system's robustness, maintainability, and usability, while also incorporating new artificial intelligence (AI) features to enhance the platform's recording analysis capabilities.

Project Objectives

The primary objectives of this project are:

- Resolve existing system bugs and restore system functionalities.
- Improve user experience and system usability.
- Improve system maintainability and support future scalability.
- Extend the capabilities of existing pages with AI features.

Features and Tasks

1. Playback and Subtitle Handling

- **Objective:** Ensure subtitles display correctly on the playback page.
- **Tasks:**
 - Fix subtitle overflow issues when a video finishes playing.

2. Admin Panel Enhancements

- **Objective:** Restore full functionality and enhance the Admin Panel.
- **Tasks:**
 - **User Tab:**
 - Implement user creation, deletion, and management.
 - Implement and enable updates of user passwords and roles.
 - **Recordings Tab:**
 - Implement sorting of recordings by upload date.
 - Allow permanent deletion of recordings.
 - **ASR Requests Tab:**
 - Restore full page functionality.
 - **Statistics Tab:**
 - Restore full page functionality.

3. Studio Page Enhancements

- **Objective:** Improve usability and data management on the studio page.
- **Tasks:**
 - Fix problems when adding tags to the studio page.
 - Display the total number of speakers.
 - Allow modification of speaker tags and save updates to the database.
 - Display additional tags on the transcript viewing page.

4. System and Infrastructure Upgrades

- **Objective:** Improve system stability and maintainability.
- **Tasks:**
 - Upgrade Node.js to version 18.
 - Resolve dependency issues caused by version upgrade.
 - Fix Docker setup issues.
 - Correctly link Docker volumes.

5. AI Transcript Summarization and Chatbot

- **Objective:** Integrate AI-powered features to enhance transcript interactions.
- **Tasks:**
 - Create a page new tab or section for transcript summarization.
 - Implement chatbot interactions to assist users in analyzing recording content.

Project Schedule

Index	Task Name	Start Date	End Date
Preparation			
1	Project Discussion and Setup	12 January 2026	26 January 2026
Implementation			
2	Playback and Subtitle Handling	27 January 2026	9 February 2026
3	Admin Panel Enhancements	10 February 2026	13 April 2026
4	Studio Page Enhancements	14 April 2026	11 May 2026
5	System and Infrastructure Upgrades	12 May 2026	22 June 2026
6	Summarization and Chatbot Features	23 June 2026	28 September 2026
Report Submission			
7	Project Plan	12 January 2026	2 February 2026
8	Interim Report	3 February 2026	24 August 2026
9	Final Report	25 August 2026	19 October 2026
10	Presentation	20 October 2026	13 November 2026

Gantt Chart

